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Role and management of light spectrum in vegetables

Gurukiran Shreemant Dathe[1*] and Shivanand Hongal[2]

[1]PG scholar, Department of Vegetable Science, College of Horticulture, Bengaluru
[2]Senior scientist and Head ICAR - KVK, Kolar

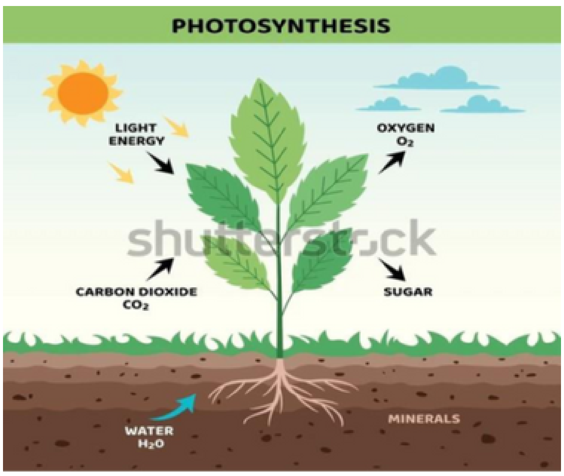
*Corresponding author email- Id: **gurukiran.4999@gmail.com**

Abstract

Light is the region of the electromagnetic spectrum that is able to create visual sensation. Light is electromagnetic radiation that can be detected by the human eye. Electromagnetic radiation occurs over an extremely wide range of wavelengths, from gamma rays with wavelengths less than about 1×10^{-11} metres to radio waves measured in metres.

Importance of light in plant growth and development
Photosynthesis:

- Light provides the energy for converting CO_2 and water into glucose and oxygen.
- Red and blue light are crucial for initiating light dependent reactions, producing ATP and NADPH for the Calvin cycle.
- Light intensity, quality and duration affect the rate of photosynthesis, influencing carbohydrate synthesis, growth and yield.
- Different wavelengths reach various canopy depths, enabling even shaded leaves to participate in photosynthesis.



Seed germination

- Light acts as a signal, not an energy source, guiding seed sprouting. Phytochromes detect red/far-red light.
- Red light promotes germination by activating phytochromes → increases gibberellins, reduces abscisic acid.
- Far-red light maintains dormancy.
- Ensures germination near the soil surface, where light is available for early growth.

Stem elongation

- Light controls stem growth via quality, intensity and duration.
- Low light or far-red dominance → triggers shade-avoidance → excessive elongation.
- Blue light (via cryptochromes/phototropins) inhibits elongation, promotes compact growth.
- Adequate light prevents weak, spindly growth by supporting balanced development.

Stomatal opening

- Blue light activates phototropism in guard cells → stomata open (via K^+ and water uptake).
- Red light indirectly aids opening by promoting photosynthesis, reducing internal CO_2.
- Ensures CO_2 uptake for photosynthesis and maintains water-use efficiency.

Flowering and reproduction

- Controlled by photoperiodism *via* phytochromes (red/far-red) and cryptochromes (blue). Short-day plants flower in long nights; long-day plants in short nights.
- Red/far-red ratio affects flowering genes (e.g., FT gene).
- Light also supports flower development, pollinator attraction, and seed set.

Secondary Metabolism

- Light regulates the production of flavonoids, anthocyanins, carotenoids and other protective compounds.
- Blue and UV light strongly induce pigmentation, antioxidant activity, and UV protection.
- Red/far-red light influences stress responses and metabolite accumulation.
- Higher light intensity/duration increases metabolite synthesis, boosting nutritional value and plant defense.

Role of different wavelengths and their function in plant growth and development

Ultraviolet (UV) Light (<400 nm)

UV light is not directly used for photosynthesis but has an important role in regulating secondary metabolism. It stimulates the production of protective compounds such as flavonoids and anthocyanins, which enhance stress tolerance, disease resistance and the nutritional quality of vegetables. UV light can also improve pigmentation and flavor, making vegetables more appealing and health-promoting.

Blue Light (400-500 nm)

Blue light plays a crucial role in regulating vegetative growth and morphology in vegetables. It promotes chlorophyll synthesis and enhances photosynthetic efficiency, ensuring robust energy production. Blue light also regulates stomatal opening, which facilitates CO_2 uptake and optimizes water use. Blue light further stimulates the production of secondary metabolites such as anthocyanins and flavonoids, which improve the color, flavor, and nutritional quality of vegetables.

Green Light (500-570 nm)

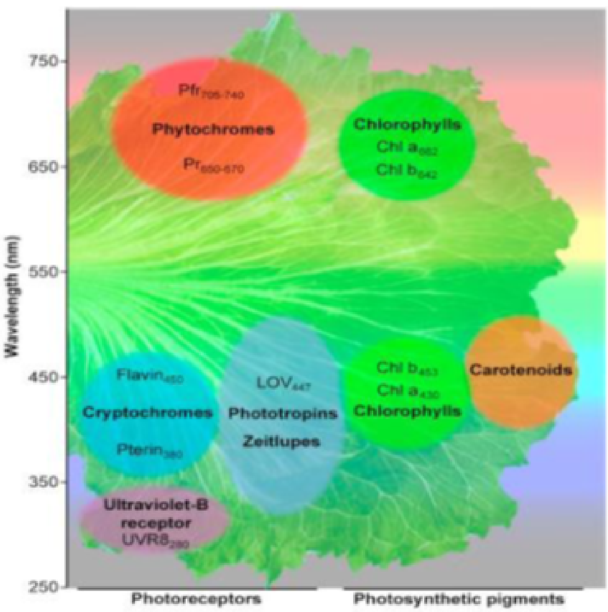
Green light, although less absorbed by chlorophyll, penetrates deeper into the plant canopy and reaches lower leaves that receive less direct light. This penetration helps maintain photosynthesis in shaded leaves and contributes to overall plant growth. Green light also plays a role in leaf expansion and canopy morphology, helping plants develop a balanced structure.

Red Light (600-700 nm)

Red light is highly effective in driving photosynthesis and promoting biomass accumulation in vegetables. It stimulates stem elongation, leaf expansion and flowering, making it essential for both vegetative and reproductive growth. Red light acts through phytochromes to regulate photomorphogenic responses such as seed germination and flowering induction.

Far-Red Light (700-750 nm)

Far-red light works in conjunction with red light via phytochromes to influence key developmental processes. It plays a major role in controlling seed germination, flowering time, and shade-avoidance responses. Under conditions where far-red light is dominant, plants elongate their stems to compete for light, which is particularly important.



Management of Light Spectrum in Vegetables

LED Lighting: LED lights are highly effective in vegetable production as they allow precise control over light quality, intensity and duration. Unlike traditional lighting, LEDs can emit specific wavelengths such as blue, red, far-red, green, or UV either individually or in combinations to trigger targeted plant responses.

Customizable spectra → growers can select R:B ratios or add FR/UV channels.

• Example: 7R:3B ratio is optimal for cucumber seedlings

Light films and covers:
Light films and covers help manage the light spectrum in vegetable production by selectively filtering, diffusing, or modifying sunlight to optimize plant growth and quality. These materials can block or reduce harmful ultraviolet (UV) radiation while enhancing beneficial wavelengths such as red and blue light, which are crucial for photosynthesis, stem and leaf development and flowering.

Supplemental and Interlighting:
Supplemental lighting and Interlighting are powerful tools for managing the light spectrum in vegetable production, especially in greenhouses and vertical farms. Supplemental lighting provides additional light from above the canopy, ensuring that plants receive adequate intensity and the desired spectral quality during low-light periods, such as cloudy days or winter seasons. Interlighting, where LEDs are installed within the canopy between plant layers, delivers targeted wavelengths directly to middle and lower leaves that are often shaded, improving photosynthesis throughout the plant.


Top lighting: Provides uniform canopy illumination (used for lettuce, spinach).
Interlighting: Placing light sources between rows in tall crops like tomato → enhances lower canopy photosynthesis and yield.



Dynamic Light Management (DLM Strategy): Dynamic Light Management (DLM) helps optimize vegetable growth by adjusting light quality, intensity and duration in real time according to the plants' developmental stage, environmental conditions, and energy needs. Unlike static lighting systems, DLM uses sensors and programmable LED arrays to deliver specific wavelengths such as blue for compact growth, red for stem elongation and flowering, or UV for secondary metabolite production exactly when and where plants require them.

Adjusting spectrum by crop stage:

Blue → Early seedling stage (compact, strong seedlings).

Red+Blue → Vegetative growth.

Red+Far Red → Flowering and Fruiting

Conclusion

The light spectrum plays a pivotal role in the growth, development and quality of vegetables by influencing photosynthesis, photomorphogenesis, flowering, fruiting and the synthesis of secondary metabolites such as flavonoids and anthocyanins Effective management of light using modern technologies, including LEDs, light films and covers supplemental and Interlighting and dynamic light management, allows precise control over intensity, duration, and spectral composition This enables growers to optimize canopy architecture, improve light penetration, accelerate growth cycles, enhance nutritional and sensory qualities and increase yield.

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